

Reading the Evidence or Reading the Country

An Identification Strategy for AI-Assisted Coding of V-Dem Democracy Indicators — Preregistration

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Table of contents

1	Study Information	2
1.1	Title	2
1.2	Research Questions	2
1.3	Hypotheses	2
2	Design Plan	6
2.1	Study Type	6
2.2	Blinding	6
2.3	Study Design	6
2.4	Randomization	8
3	Sampling Plan	8
3.1	Existing Data	8
3.2	Explanation of Existing Data	8
3.3	Data Collection Procedures	9
3.4	Sample Size	9
3.5	Sample Size Rationale	10
3.6	Stopping Rule	10
4	Variables	10
4.1	Manipulated Variables	10
4.2	Measured Variables	10
4.3	Indices	12
5	Analysis Plan	12
5.1	Statistical Models	12
5.2	Transformations	12
5.3	Inference Criteria	12
5.4	Data Exclusion	13
5.5	Missing Data	13
5.6	Exploratory Analysis	14

1 Study Information

1.1 Title

Reading the Evidence or Reading the Country: An Identification Strategy for AI-Assisted Coding of V-Dem Democracy Indicators

1.2 Research Questions

Large-scale expert annotation projects like the Varieties of Democracy (V-Dem) indicators depend on panels of country experts whose judgment is expensive to sustain and difficult to replace. Median panel size across V-Dem indicators fell from roughly eleven coders in the 2010s to 5.8 by 2024 as the original cohort of experts aged out faster than it was replaced. Large language models (LLMs) are an obvious candidate for augmenting thinning expert panels, but only if their ratings reflect genuine evaluation of the evidence in front of them (contextual knowledge) rather than information encoded during pretraining (parametric knowledge). A model that has memorized a country’s general reputation can appear well-calibrated without having the capacity to process new information due to the tendency of LLMs to heavily rely on their pretrained weights. This distinction determines whether AI ratings are safe to rely on for cases featured less prominently featured in public discourse or those undergoing rapid transformations like regime change. This study addresses four interrelated questions:

1. When a large language model rates a country’s political conditions on a V-Dem indicator, is it drawing on the textual evidence it has been given in the prompt, or on prior beliefs about the country activated by its name or other identifying details in the text?
2. Does embedding calibration in the model’s weights through fine-tuning outperform supplying few-shot calibration examples in the prompt, and does training on de-identified (anonymized or summarized) text change what the model learns to rely on?
3. Do the answers to these two questions, established on a single validation year, hold up when the best-performing model is carried forward to later years, including a year entirely outside the model’s training data, and under what conditions, such as a country undergoing a regime transition, does the model rely more or less heavily on the text in front of it?
4. Is the resulting model’s agreement with the human expert panel good enough that adding its ratings to a thinning panel would not measurably shift the panel’s consensus?

1.3 Hypotheses

The primary metric throughout is the artificial intelligence mean absolute error (AI MAE), the absolute deviation of the AI rating from the raw human panel mean for a given country-year-indicator (CYI), bootstrapped at that level. Our explicit assumptions here are (a) that the textual sources in the prompts provide useful information to the AI and (b) that a well-calibrated AI that pays attention to important changes highlighted in the textual sources will better track the ratings of a panel of human experts.

Four prompt conditions, named below and defined fully in the Design Plan, run through every part of this section. **Codebook** gives the model the indicator’s definition and response categories only, no source text and no calibration examples, and measures whatever calibration the model already

carries from pretraining. **Evidence** adds the relevant sections of the State Department and Freedom House reports, along with calibration examples showing panel-mean ratings for other countries on the same indicator. **Anonymized** repeats the evidence condition with the country’s name and other identifying details stripped from both the evidence text and the calibration examples. **Summarized** goes further still, replacing the evidence and calibration examples with a short, generic rewrite of the same information that strips not just names but the kind of institutional detail that can still give a country away. For each of the three textual conditions (raw evidence, anonymized text and summarized text) we explore differences in model performance across few-shot example prompts and calibrating model weights directly through QLoRA fine-tuning.

The confirmatory hypotheses below are organized into four parts that mirror the four research questions above. We provide the directional reading of the main hypothesis along with alternative hypotheses where relevant. The full operationalization, including the equivalence band used to interpret small MAE differences, is given in the Analysis Plan. A further set of exploratory hypotheses, reported with the same bootstrapped confidence intervals but not treated as pass or fail tests of the study’s central claims, is also described in that section.

Part 1. Base-model identification (2019 data). Four progressively de-identified prompt conditions (codebook-only, raw evidence, anonymized evidence, summarized evidence) are used to test whether the base model is drawing on source text or on country-identity priors.

- **B1.** *Evidence beats codebook-only: $MAE(Evidence) < MAE(Codebook)$.* If the model is taking into account the evidence packets, then the introduction of evidence packets and calibration examples should improve their ability to reproduce the ratings of human expert panels.
 - **B1a.** The model focuses too narrowly on the text in a way that distracts it from an informative prior, resulting in *worse* performance. A second interpretation is that the model may be using the evidence but weighting or interpreting it differently than the human panel, producing information-driven divergence rather than non-use.
- **B2.** *The information gain from the evidence packets survives anonymization: $MAE(Anonymized) < MAE(Codebook)$.* If the model can extract real calibration signal from text alone, with zero country-identity information anywhere in the prompt, then anonymized evidence and calibration examples should beat a codebook-only baseline that has the country name but no text at all. This result helps to rule out country name recall as the sole driver of improvement.
 - **B2a.** The country name and related information from training are doing most of the work, while de-identified text can’t make up for losing it, such that anonymized performs *worse* than codebook-only.
- **B3.** *Anonymization beats raw evidence: $MAE(Anonymized) < MAE(Evidence)$.* If evidence packets activate parametric knowledge in a way that pulls ratings toward the country’s general reputation rather than what the specific text describes, then removing details about named entities should improve the model’s ability to reproduce human panel ratings for a specific CYI.
 - **B3a.** The parametric knowledge is *net beneficial* and supplements thin or ambiguous text rather than distorting it. A second interpretation is that the anonymization pass itself damaged evaluative content, since an LLM rewrite is not a surgical redaction and may discard genuinely evaluative detail independent of whether the identity itself carried any calibration value.

- **B4.** *Summarization reduces residual identity leakage and improves on anonymized: $MAE(\text{Summarized}) < MAE(\text{Anonymized})$.* Preliminary re-identification testing shows anonymized text remains substantially reidentifiable (~51–61% top-1, vs. a <30% target for summarized), which motivated adding summarization to the design. If that residual identifiability is activating parametric knowledge, then summarization’s sharper de-identification should further reduce reliance on parametric knowledge and improve model calibration over anonymized text.
 - **B4a.** The abstraction cost of summarization outweighs the leakage-reduction benefit and summarized performs *worse* than anonymized because the generic rewrite discards genuinely evaluative content faster than it removes usable identity leakage. A second interpretation is that the summarized text still carries adequate signal, but the model weights or interprets that residual signal differently than the human panel does, producing divergence that reflects genuine information use rather than net content loss.

Part 2. Fine-tuning (2019 data). Three fine-tuned variants of the base model, each trained on one of the three evidence types, are used to ask whether embedding calibration in the model’s weights outperforms in-context calibration examples, and whether training on de-identified text changes what the model attends to.

- **F1.** *Fine-tuning beats few-shot prompting under matched conditions: $MAE(FT\text{-}Raw) < MAE(Base\text{-}Raw)$, $MAE(FT\text{-}Anonymized) < MAE(Base\text{-}Anonymized)$, and $MAE(FT\text{-}Summarized) < MAE(Base\text{-}Summarized)$.* The rationale is that the larger number (up to 100K training examples) embed more calibration signal than five prompt exemplars can. The comparison deliberately excludes few-shot weights in favor of fine-tuned weights to enable the comparison of prompt-based calibration versus weight-based calibration.
 - **F1a.** Fine-tuning introduces overfitting, thereby reducing performance relative to the base model on the held-out sample.
- **F2.** *Model performance on reproducing human ratings is conditioned by the extent to which they reduce re-identification bias, e.g. $MAE(FT\text{-}summ:Evidence) < MAE(FT\text{-}anon:Evidence) < MAE(FT\text{-}raw:Evidence)$.* If the text is providing useful information, then identity leakage resulting from named entities in the prompt or text will hurt model performance by inducing it to rely on pretrained knowledge rather than the textual evidence presented in the evidence packets. Anonymized evidence and further abstraction through summarization should therefore improve the ability of models to reproduce expert scores by focusing the model’s attention on important textual details. We choose the raw evidence condition as the basis for the test, since it is the harder test relative to evaluating each model on the condition it was trained on. Only FT-raw trained on this exact distribution, so FT-anon and FT-summ face a distribution mismatch the ranking has to overcome.
 - **F2a.** A different ordering could yield three potential interpretations. First, FT-raw is trained on raw text and therefore has an inherent advantage on reading and processing the raw text. A second possibility is that the abstraction cost of summarization outweighs the leakage-reduction benefit such that abstraction away from the raw text induces progressively *worse* performance, e.g. the anonymized and summarized text discard genuinely evaluative content more than they remove identity leakage. A third, training-side interpretation is that fine-tuning on progressively de-identified text may teach the model to attend to different features of the input than fine-tuning on raw text does.

Part 3. Robustness and mechanism tests (best model on 2023/2024 data). The best-performing model from Parts 1 and 2 is carried forward to later years and put through a series of mechanism tests designed to isolate how textual information actually influences its ratings, independent of whether it improves fidelity to the panel mean.

- **R1.** *Main results replication.* Replication of primary results will hold with 2023 data: $\text{MAE}(\text{Evidence}) < \text{MAE}(\text{Codebook})$, $\text{MAE}(\text{Anonymized}) < \text{MAE}(\text{Codebook})$, $\text{MAE}(\text{Anonymized}) < \text{MAE}(\text{Evidence})$, and $\text{MAE}(\text{Summarized}) < \text{MAE}(\text{Anonymized})$. The main results for the best model are recomputed for 2023 and compared to the 2019 estimates. If the identification results replicate in direction and magnitude, the decomposition tests generalize beyond the validation year.
- **R2.** *Regime transitions test.* The evidence gain is larger for countries undergoing a regime transition. The gap between the evidence condition and the codebook condition $\text{MAE}(\text{Evidence}) - \text{MAE}(\text{Codebook})$, and the same for Anonymized and Summarized will be more negative for ERT-tagged transition-adjacent country-years than for stable ones. The carried model’s frozen prior (whatever it learned during pretraining or fine-tuning) describes the country as it was at some point at or before 2019. In transition-adjacent years, that snapshot has fallen out of date with 2023 reality, so the evidence text that describes 2023 conditions directly should carry more marginal information than in stable years, where the frozen snapshot and the current text roughly agree.
- **R3.** *2024 holdout test.* The evidence gain over codebook-only persists, or grows, in 2024 (the year after Llama 3.3’s December 2023 knowledge cutoff). In 2024, input leakage (e.g. leakage of the textual evidence) is impossible given that the model was not trained on 2024 data. Thus, any difference between codebook only and the other conditions must be due to the presentation of new textual evidence. Since State Department reports change in format and substance in 2024, we run this test using Freedom House data only. The test includes the carried model across all four conditions with 2024 FH-only compared with a 2023 FH-only companion.
- **R4.** *Name-swap test.* Name-swap ratings track described conditions, not the named country. We take within-region country pairs and swap the names attached to the evidence packets, then ask the model to rate the country on the specified indicator. If the textual information matters, the model will base its ratings on the textual information and not the country name alone. We therefore expect the ratings of the swapped pairs to track more towards the panel mean of the actual source country described by the documents rather than the named country, e.g. $\text{MAE}(\text{Source}) < \text{MAE}(\text{Named})$, tested with a paired comparison, differencing within-case before resampling.
 - **R4a.** $\text{MAE}(\text{Named}) < \text{MAE}(\text{Source})$. Ratings anchor to the named country’s identity rather than the described conditions, i.e., country-identity priors override the text even when they directly conflict with it.
- **R5.** *Re-identification test.* We use re-identification success on anonymized and summarized text as a salience filter. We expect the gap between $\text{MAE}(\text{Codebook})$ and $\text{MAE}(\text{Evidence})$ to be smaller for re-identified (salient) cases than for non-identified (non-salient) cases, because a strong competing prior lets the model discount the text for salient cases, while non-salient cases leave it more reliant on what the text actually says.
 - **R5a.** The gap between $\text{MAE}(\text{Codebook})$ and $\text{MAE}(\text{Evidence})$ is larger for salient cases than for non-salient cases: evidence helps salient cases more, not less. This could happen

if salient countries also have richer source documentation, and that richness outweighs the competing prior.

Part 4. Deployment. Having established what the model is drawing on and how well that generalizes, the final question is whether the resulting ratings are good enough to use.

- **D1. Agreement test.** The carried-forward model’s AI MAE, in its best evidence condition, is at or below the human leave-one-out (LOO) MAE. This will be verified with a 2019 diagnostic via a human LOO reference line in Figure 1 and a confirmatory test on 2023 data per the design doc’s Agreement test. If AI MAE > human LOO MAE, the model deviates from panel consensus by no more than a typical human coder.
- **D2. Thin-panel augmentation test.** Adding one AI rating to a thin panel (8 coders, 2023) does not shift the panel mean beyond normal human-coder replacement variance: the augmentation divergence stays below the empirically-derived 90th-percentile threshold computed from human-only coder swaps in the same pool.

2 Design Plan

2.1 Study Type

This is a fully-crossed factorial experiment conducted on model behavior rather than on human subjects. The research team manipulates two things, which evidence a model is given in the prompt (the four conditions below, plus three zero-shot variants), and, for three of the four models, what kind of text the model was trained on. Those manipulations are applied to a fixed, existing population of country-year-indicator cells drawn from V-Dem’s coder-level data, not to a randomly sampled or randomly assigned set of human participants.

2.2 Blinding

Blinding in the human-subjects sense does not apply, since there are no human participants in the study itself. The relevant analogue is that the benchmark the AI ratings are scored against, the human panel mean for each country-year-indicator, was fixed by V-Dem’s coder panels well before this study began and before any AI rating was produced. No AI output could have influenced the human ratings, and the scoring procedure (absolute deviation from that pre-existing panel mean) leaves no discretion for a researcher to shape a result after seeing it.

2.3 Study Design

Four prompt conditions test what the model draws on when it rates a country-year-indicator:

- *Codebook-only* gives the model the indicator definition and response categories but no source text and no examples, which measures whatever calibration the model already carries from pretraining.
- *Evidence* adds relevant sections of the State Department’s Country Reports on Human Rights and Freedom House’s Freedom in the World report, along with five calibration examples showing panel-mean ratings for other countries on the same indicator.

- *Anonymized* repeats the evidence condition with the country’s name and named entities such as officials and institutions, and datable events, stripped from both the evidence text and the calibration examples.
- *Summarized* goes a step further. Instead of redacting names from the original text, a separate model call rewrites the assembled evidence as a short, generic, indicator-targeted description of political conditions, discarding not just names but the kind of institutional detail that can still give a country away even after its name is removed.

The base model is also run on the evidence, anonymized, and summarized conditions with the calibration examples removed, to separate what the source text itself contributes from what the calibration examples contribute (few-shot ablation).

Four models are compared under these conditions. The base model is Llama 3.3 70B Instruct, used as released. Three additional variants are produced by fine-tuning that same base model, each on a different kind of text: FT-raw on raw evidence text; FT-anon on anonymized text; and FT-summ on summarized text.

Fine-tuning uses QLoRA, a parameter-efficient method (low-rank adaptation) that trains a small number of additional weights, rather than updating the full model (the exact training hyperparameters are recorded in the replication package). All three variants train on V-Dem coder-level ratings from 2016 through 2018, a window chosen because source documents are reliably available from 2016 onward and because it precedes both the 2019 evaluation year and the later robustness years, so there is no overlap between what a model trains on and what it is scored against.

Training pairs each individual coder’s rating, not the panel mean, with the corresponding section text, so a country-year-indicator rated by several coders contributes several training examples with the same input text and different target ratings. Full-scale training on the entire pool of roughly 900,000 coder-level examples is not computationally feasible within the study’s timeline, so each variant instead trains on a shared subsample of about 100,000 cases. The three variants train on the same underlying set of cases (drawn from the cases available in common to all three text versions), allocated proportionally across indicators with a minimum number of examples reserved for each rating level so that rare ratings are not left out entirely.

Because the raw-versus-anonymized-versus-summarized comparison depends on the three variants seeing the identical set of cases, a case is dropped from all three training sets if its assembled prompt runs too long for even one of the variants to process, rather than letting each variant keep or drop cases independently. Training stops early once performance on a held-out validation set of cases stops improving, rather than running a fixed number of passes over the data. That held-out slice is built by withholding whole country-year-indicator cells rather than individual coder rows, because several coders’ rows on the same cell share a nearly identical prompt and a row-level split would let the model see, and effectively memorize, prompts that also appear in its own validation set.

At inference, the fine-tuned models are queried without any calibration examples in the prompt, since calibration is meant to live in the trained weights instead. Every model, at every stage, is queried at a fixed decoding setting that always returns its single most likely rating rather than a randomly sampled one, so a repeated call on the same input returns the same rating.

The primary analysis uses 2019 as the evaluation year, chosen as a clean one-year gap after the fine-tuning window with full coder panels and no major exogenous disruption. Robustness and mechanism tests, described further in the Analysis Plan, are then run on 2023 data and on 2024

data restricted to Freedom House sources, using whichever of the four models performs best in the primary analysis.

2.4 Randomization

There are no human participants to randomly assign, and randomization plays no role in inference. Every model is run over the full universe of eligible country-year-indicator cells rather than a sampled subset, and each rating is produced deterministically rather than drawn from the model’s output distribution. Randomization enters only in building the fine-tuning data, where a fixed seed governs which cases are drawn into each variant’s shared subsample, and separate fixed seeds govern the shuffling of training examples and the initialization of the fine-tuned weights during training. These seeds are fixed in advance and recorded in the replication package, so the same draws can be reproduced exactly.

3 Sampling Plan

3.1 Existing Data

This study uses data that already existed before the study was conceived. V-Dem’s individual coder-level ratings (version 15) and the source documents behind the evidence packets, including the State Department’s Country Reports on Human Rights and International Religious Freedom Reports and Freedom House’s Freedom in the World reports, were produced independently of this project. The research team has already used parts of this existing data, mainly the 2016–2018 training window, the codebook text, and a small number of diagnostic samples described below, to build the pipeline and lock the design decisions above. No AI rating has yet been examined against the confirmatory 2019, 2023, or 2024 evaluation pools that the hypotheses in this document will be tested against, and no analysis of the kind described in the Analysis Plan below has been run.

3.2 Explanation of Existing Data

Four pieces of prior work touched this data before the design was locked, and are disclosed here rather than folded silently into the results.

First, a 98-country-year-indicator pilot sample from 2019 was used to compare how easily a country could be guessed back from anonymized text with and without the executive summary included. That pilot is what motivated the policy of only falling back to executive-summary text when no body section is available, and it is what motivated adding the summarized condition to the design in the first place, since it showed that name-redaction alone left a country guessable roughly half to two-thirds of the time. Because this pilot used real 2019 evidence text and real model calls on a small slice of the primary evaluation year, those 98 country-year-indicator cells are excluded from the confirmatory 2019 evaluation pool described below, so that no cell contributing to the confirmatory hypothesis tests was ever part of a decision that shaped the design.

Second, one full fine-tuning run was completed for each of the three variants and then discarded after the team discovered that the training script’s internal validation split had been drawn at the individual coder-row level rather than the country-year-indicator level, which let near-duplicate

prompts appear on both sides of the split and made the early-stopping signal unreliable. No inference and no evaluation was ever run on those discarded adapters; the only thing examined from them was their training-loss curves, used to diagnose the split defect itself. All three variants are being retrained from scratch under the corrected, cell-level split described in the Design Plan.

Third, an early two-condition smoke test on a smaller Llama model was run to confirm the pipeline could produce well-formed output. Its results were never examined for content and played no role in any design decision.

Fourth, inference runs for the confirmatory 2019, 2023, and 2024 evaluation pools have already been submitted on the HPC cluster, ahead of finalizing this document, to avoid losing queue position given potential scheduling bottlenecks on the cluster. No output from these runs has been downloaded from the cluster, inspected, or compared to any panel mean; the ratings exist only as unretrieved job output and have played no role in any design decision in this document.

Beyond these four items, the only other prior use of the existing data has been ordinary preparation work with no bearing on results. This includes confirming that source documents for the 2019 and 2023 evaluation years were fully downloaded and free of parsing errors, and building the indicator-to-section mapping and the calibration example set from the 2016–2018 training window, none of which involved comparing an AI rating to a panel mean in any evaluation year.

3.3 Data Collection Procedures

No new data is collected from human subjects. The evaluation pool for a given year is assembled programmatically rather than sampled by hand. For every indicator in the 206-indicator configuration file, every country with a raw panel mean recorded for that indicator and year is included, provided a processed source document exists in both the State Department and Freedom House archives for that country-year. This produces the full population of eligible cells for that year, and every model and condition is run against that same population. Nothing is held back or subset by the research team after the fact.

Of the 206 indicators in the configuration file, 205 have any panel-mean data to evaluate against in any year from 2010 through 2024; the remaining indicator (`v2exl_legitideolcr`) was retained in the pipeline for section-mapping completeness but has no panel ratings in any year and so never enters the evaluation pool.

3.4 Sample Size

The primary 2019 evaluation pool covers the full universe of roughly 205 evaluable indicators across all eligible country-years, on the order of 32,800 country-year-indicator cells per prompt condition per model. The 2023 robustness pool is comparably sized, and the 2024 Freedom-House-only structural holdout is somewhat smaller, reflecting both the restriction to a single source and continued attrition in panel sizes. No minimum number of coders is required for a cell to enter the primary evaluation pool as any cell with at least one coder rating is included. A floor of at least two coders applies only to the human leave-one-out benchmark used in the agreement test, since that quantity is mathematically undefined when there is only one coder to remove. Cells below three coders make up a small fraction of either pool (well under one percent in both 2019 and 2023), so this floor has little practical effect on the pool as a whole.

3.5 Sample Size Rationale

The evaluation pool is the full population of eligible cells rather than a sample drawn from a larger one, so there is no sampling procedure to justify in the usual sense. An earlier version of the design considered a stratified fallback sample, to be used only if a much larger model (405 billion parameters) could not be run to completion within a limited number of compute allocations. That contingency no longer applies, since the larger model was dropped from the design outright for reasons unrelated to compute availability on the day of any given run, so there is no remaining trigger that would call for a fallback sample. Running the full population instead of a sample also avoids any question of whether a particular stratification could have been chosen after seeing preliminary results.

3.6 Stopping Rule

Each of the three fine-tuned variants trains until its performance on the held-out slice of cases described in the Design Plan stops improving by a meaningful margin over ten consecutive evaluation checkpoints, spaced 100 training steps apart. Once that point is reached, training halts immediately, and the checkpoint kept is whichever one had the lowest held-out loss, not necessarily the last one produced. If a variant instead completes a full pass over the training data with its held-out loss still improving, training is extended by one additional pass over the same subsample rather than stopped outright, up to a fixed cap of three passes. This rule is applied identically and automatically to all three variants, so none of them ends up with a longer or shorter training run because of how its results looked partway through.

Inference has no stopping rule of its own. Each batch runs the full eligible population of country-year-indicator cells for a given year, condition, and model to completion, so there is no partial sample or interim result that could cut a run short.

4 Variables

4.1 Manipulated Variables

Two things are varied by design. The first is the prompt condition: codebook-only, evidence, anonymized, and summarized, along with the three zero-shot versions of the latter three. The second is which of the four models produces the rating, i.e. the base model or one of the three fine-tuned variants, each distinguished by which kind of text it was trained on. Both are described fully in the Design Plan above.

4.2 Measured Variables

The outcome variable in every confirmatory hypothesis is AI MAE, the absolute difference between the AI's rating and the raw human panel mean for a given country-year-indicator, computed as $AI_MAE = |AI_rating - panel_mean|$. This is computed once per country-year-indicator, condition, and model, and bootstrapped at the country-year-indicator level (2000 resamples) to produce confidence intervals for every comparison in this document. AI MAE is measured against the panel

mean rather than against each individual coder’s rating because the alternative, averaging the AI’s distance from every coder separately, is always at least as large as the panel-mean version, and the gap between the two is nothing more than ordinary disagreement among the human coders themselves.

It is important to note that AI MAE has no natural floor at zero. Ratings are integers, but a panel mean is typically a fraction, so even a rating that reflects the panel mean perfectly still has to round to the nearest whole number and absorb a small rounding error. The panel mean is also itself an estimate drawn from a small group of coders, and carries its own sampling noise on top of the rounding error. Since the rounding error and panel sampling noise apply to every model and condition, they do not require corrections in the context of hypothesis testing.

However, for model selection and deployment, it is important to have a baseline against which to judge the performance of the AI models. For this purpose, we compare AI MAE against a human leave-one-out (LOO) MAE baseline. For each coder in a country-year-indicator’s panel, this removes that coder from the panel, recomputes the mean of the rest, and takes the coder’s absolute distance from that reduced mean; averaging this across all coders in the panel gives a per-cell figure describing how far a typical individual coder sits from the consensus of their peers. This provides a direct comparison between the AI’s MAE and a human in a directly analogous position to the AI, i.e. rating an indicator as a non-panel member. Comparing the AI MAE and the human LOO MAE thereby shows directly whether the AI deviates from the panel consensus by more or less than a typical human coder.

Two secondary calibration summaries are reported alongside AI MAE, mainly because they are easier to read for an audience not used to thinking in absolute-error terms. The exact match rate is the share of AI ratings equal to the rounded panel mean, and adjacent-category agreement is the share of AI ratings within one category of the rounded panel mean.

A separate diagnostic, signed deviation, captures direction rather than magnitude, `AI_rating - panel_mean` without the absolute value, averaged within quintiles of a country-year’s democracy score. Individual human ratings are balanced around their own panel mean, so the human baseline for signed deviation is close to zero throughout the distribution. A systematic AI departure from zero in a particular quintile, ratings running high in the most autocratic quintile or low in the most democratic one, would indicate the AI is compressing its ratings toward the middle of the scale, or shading them in one direction depending on regime type, in a way that AI MAE alone cannot show.

Three further measured variables support the Part 3 mechanism tests specifically, defined in more detail where each test appears in the Analysis Plan below. Re-identification accuracy is the share of anonymized or summarized cases where the model correctly guesses the true country from the text alone, reported as both a top-1 and a top-3 figure. Signed deviation by re-identification status is the same signed-deviation calculation above, split by whether a case was correctly re-identified. Name-swap MAE is the AI’s distance from the named country’s panel mean against its distance from the true source country’s panel mean, in cases where the two have been deliberately mismatched.

The thin-panel augmentation test (D2) uses its own measure, divergence, the absolute change in a panel’s mean when a single AI rating is added to it, computed as `|mean(panel + AI_rating) - mean(panel)|`.

4.3 Indices

No composite indices are constructed. Every measure above is reported and tested on its own; none are combined into a summary score.

5 Analysis Plan

5.1 Statistical Models

Parts 1 and 2 are evaluated on all four models directly. Which model is carried forward into Part 3 is decided by a fixed rule rather than by judgment after seeing the Part 3 results. It is whichever of the four models has the lowest 2019 AI MAE in the raw evidence condition, with ties (overlapping bootstrap confidence intervals) resolved in favor of the base model as the simpler of the two. The Part 3 and Part 4 confirmatory hypotheses (R1 through R5, D1, D2) are registered against that carried-forward model only. Where compute allows, the same 2023 and 2024 battery is also run on the other three models, but those results are reported as exploratory cross-model comparisons rather than as tests of the confirmatory hypotheses.

Every comparison in this document, from the four base-model conditions through the mechanism tests, is evaluated with the same procedure. The relevant statistic (an AI MAE, a difference between two AI MAEs, an accuracy rate, a divergence) is computed at the country-year-indicator level, then country-year-indicators are resampled with replacement 2000 times to build a bootstrap distribution, and its 2.5th and 97.5th percentiles are taken as the 95 percent confidence interval. Where a hypothesis compares two conditions or two benchmarks on the same underlying cases, as in the name-swap test, the difference is computed within each case first and only then resampled, so the pairing is preserved rather than treating the two sides as independent samples.

5.2 Transformations

Few of the variables above require transformation before analysis. Democracy quintiles are constructed by sorting country-years into five equal-sized groups on `v2x_polyarchy`. Regime-transition status uses the V-Dem Episodes of Regime Transformation dataset's onset and peak year flags as the primary indicator, with the year-over-year change in `v2x_polyarchy` as a continuous moderator and the threshold for that continuous measure is fixed before the 2023 mechanism tests are run. Coverage tier (strong, partial, weak, or no direct section mapping) is a fixed categorical label assigned during indicator-to-section mapping, not something computed from the data being analyzed. AI MAE itself requires no transformation beyond the absolute-value calculation already described under Measured Variables.

5.3 Inference Criteria

Because AI MAE is bootstrapped across roughly 32,800 country-year-indicator cells, even a trivially small difference can produce a confidence interval that excludes zero. Statistical significance at this sample size is not the same thing as a difference large enough to matter substantively. We therefore fix a smallest effect size of interest, or SESOI, before running any confirmatory comparison. It

is set at half of that year’s rounding floor, the small, mechanical error described under Measured Variables that even a perfectly calibrated rater cannot avoid (roughly 0.115 in 2019, and similarly sized in 2023 and 2024). Any MAE difference whose 95 percent confidence interval falls entirely within that band, on either side of zero, is described as not representing a substantively meaningful effect, regardless of whether the interval technically excludes zero. This is a standard for how a result is described, rather than a rule for excluding it from the write-up, and every point estimate and confidence interval is reported in full either way.

For most hypotheses in this document, the directional, alternative, and null readings listed under Hypotheses above are decided the same way. A confidence interval that clears both zero and the SESOI band in the predicted direction supports the directional reading. One that clears both in the opposite direction supports the alternative reading. One that includes zero, or falls entirely inside the SESOI band, supports the null reading.

Two hypotheses use a different procedure because they are not simple deltas against zero. The name-swap test (R4) is a paired comparison. For each swapped case, the AI’s distance from the source country’s panel mean is subtracted from its distance from the named country’s panel mean before resampling, so the test asks whether that within-case difference clears zero and the SESOI band, in either direction, rather than comparing two unpaired MAE figures. The thin-panel augmentation test (D2) is judged against a threshold rather than against zero. Using only the human side of the 2023 thin-panel pool, we compute how much a panel’s mean moves when a single ordinary coder is removed from it, for every coder in every eligible cell, and take the 90th percentile of that distribution as the threshold, before looking at a single AI rating. The augmentation divergence from adding an AI rating is judged to exceed normal replacement tolerance only if the lower bound of its own 95 percent confidence interval clears that threshold, not merely its point estimate. The agreement test (D1) is a direct comparison against the human leave-one-out MAE benchmark rather than a delta against zero, so no SESOI band applies to it; the carried-forward model’s AI MAE is judged against that benchmark using the same bootstrap confidence interval as everywhere else.

5.4 Data Exclusion

The 98-country-year-indicator re-identification pilot described under Sampling Plan’s Existing Data section above is excluded from the confirmatory 2019 evaluation pool used to test every Part 1 and Part 2 hypothesis. Beyond that, no cell is excluded from the AI MAE analysis on the basis of what the AI produced; a rating that looks unusual is still a data point, not a candidate for removal. The only other exclusion is the coder-count floor described under Sampling Plan, which removes cells with fewer than two coders from the human leave-one-out benchmark alone, since that quantity cannot be computed with only one coder to remove. If a model’s response cannot be parsed into a valid rating even after the parser’s fallback pass, that cell is left incomplete rather than assigned a default value, and is filled in on a subsequent run of the same batch rather than dropped from the design.

5.5 Missing Data

A country-year-indicator with no raw panel mean on record is not part of the evaluation pool to begin with, as described under Sampling Plan, so there is no panel-mean value to be missing within the analysis itself. Missing source text is handled by the fallback hierarchy fixed in the

Design Plan. A cell with a direct section mapping in only one of the two source archives is coded from whichever source is available, and a cell with no direct section mapping in either archive falls back to the executive summary alone. A cell is only left uncoded if the anonymized or summarized text it needs has not yet been generated for that country-year, which is a pipeline-completeness gap rather than genuine missingness in the data itself, and is resolved by completing that processing step rather than by imputing or dropping the cell.

5.6 Exploratory Analysis

The following hypotheses are exploratory. They are reported with the same bootstrapped confidence intervals and the same SESOI band as the confirmatory hypotheses above, but are not treated as pass or fail tests of the study’s central claims.

Few-shot ablation. To tease out the effects of calibration examples relative to textual evidence, we run inference on each of the textual evidence conditions without the few-shot calibration examples.

- **A1.** Zero-shot text alone, without calibration examples, still improves on codebook-only: $\text{MAE}(\text{Evidence-zero-shot}) < \text{MAE}(\text{Codebook})$, $\text{MAE}(\text{Anonymized-zero-shot}) < \text{MAE}(\text{Codebook})$, and $\text{MAE}(\text{Summarized-zero-shot}) < \text{MAE}(\text{Codebook})$. This is the test of genuine text-reading disentangled from the calibration-block confound.
 - **A1a.** Zero-shot text performs worse than codebook in one or more conditions, suggesting that without calibration examples to anchor interpretation, added text actively confuses the model relative to having no text at all.
- **A2.** We expect that the marginal value of the few-shot calibration examples will be conditioned by how much identity-based anchoring is available. The few-shot calibration gap for each condition, $\text{MAE}(\text{condition, few-shot}) - \text{MAE}(\text{condition, zero-shot})$, should be close to zero for Evidence, and increasingly negative for Anonymized and Summarized. If country identity already provides the model with a scale anchor based on its pre-training, few-shot examples should contribute little to performance, but as anonymization and summarization progressively strip those anchors, the model should lean more on the calibration block to map described conditions onto the rating scales.
 - **A2a.** The gradient reverses such the calibration gap is largest for evidence and shrinks through anonymized and summarized, implying identity and content cues make the model *more* reliant on calibration examples.

Additional fine-tuning robustness.

- **A3.** Raw fine-tuning bakes in identity anchoring (the codebook probe): $\text{MAE}(\text{FT-raw:Codebook}) < \text{MAE}(\text{FT-anon:Codebook}) < \text{MAE}(\text{FT-summ:Codebook})$. We run inference using the fine-tuned models on the codebook-only condition (no textual evidence provided in prompts). If FT-Raw’s number is much better than the FT-Anonymized or FT-Summarized models then it must be something FT-raw’s weights learned during training, e.g. an association between the real country name and its typical rating, baked in because FT-raw’s training data always paired real names with true ratings. We further anticipate that anonymized training text leaks identity more often than summarized, giving the model more incidental chances to learn a country-conditioned shortcut anyway.

- **A3a.** The gradient collapses to a flat step, FT-anon:Codebook – FT-summ:Codebook, both well behind FT-raw:Codebook, meaning incidental re-identification during FT-anon training wasn’t frequent enough to teach a shortcut; only explicit name-rating pairing (FT-raw) does.
- **A4.** Training-side analog of B4, operationalized as a difference-in-differences: $DiD = [MAE(FT\text{-}summ:Summarized) - MAE(FT\text{-}anon:Anonymized)] - [MAE(Base:Summarized) - MAE(Base:Anonymized)]$ (the second bracketed term is B4’s own delta). Bootstrapped at the CYI level. The training-time tradeoff replicates the inference-time tradeoff from B4 if DiD’s CI falls within the equivalence band defined above (50% of that year’s rounding floor on either side of zero, roughly 0.115 in 2019).
 - **A4a.** DiD’s CI falls entirely outside the equivalence band: training and inference diverge. Fine-tuning on summarized text either erases much more of the tradeoff than reading it at inference did, or makes it much worse, meaning the training process itself changes how much the de-identification/specificity tradeoff matters, rather than just inheriting it from the text.
- **A5.** Training-representation transfer to raw text. FT-anon and FT-summ, despite never training on raw evidence, still beat the base model when reading raw evidence at deployment: $MAE(FT\text{-}anon:Evidence) < MAE(Base:Evidence)$ and $MAE(FT\text{-}summ:Evidence) < MAE(Base:Evidence)$. This is because the calibration and text-reading ability learned from de-identified text carries over to raw text, which represents a superset of what they trained on.
 - **A5a.** The representation mismatch dominates instead: FT-anon and/or FT-summ perform worse than Base on raw evidence, because real names and other identity cues at test time are inputs the model never learned to handle, and the mismatch outweighs whatever generic calibration fine-tuning provided.

Additional re-identification robustness.

- **A6.** *Re-identification diagnostic.* We ask the model to try to re-identify the countries in anonymized and summarized text. We expect that summarization achieves materially lower re-identification rate than anonymization because summarization abstracts further from the original text than anonymization.
 - **A6a.** Summarization achieves a higher re-identification rate than anonymization, suggesting that distilling text down to only the most distinctive traits can make a country easier to guess, not harder.
- **A7.** *Re-identification predicts directional bias.* In the anonymized and summarized conditions, cases where the model correctly re-identifies the country show larger directional (signed) deviation from the panel mean than non-identified cases (too high for autocracies, too low for democracies), consistent with anchoring on a recovered country prior. Reported by regime type and by region.
- **A8.** *Re-identification: salience gap by model.* By the same logic, de-identified training should blunt this salience effect. FT-raw should replicate the base model’s salience gap, since its weights learned to pair real country names with ratings. FT-anon and FT-summ, whose training data never paired a real name with a rating, should show a smaller gap between salient and non-salient cases.

Coverage.

- **A9.** *Source coverage test (2023).* Calibration degrades with weaker source coverage: AI MAE is higher for weak-coverage indicators than for strong-coverage indicators, across all conditions, for the carried-forward model. Other models are covered only where the exploratory multi-model extension (Part 3 intro) actually runs them.

A further set of appendix-level analyses uses output already generated for the analyses above, requires no additional inference, and is reported descriptively rather than as hypothesis tests. These include whether the evidence condition still helps on indicators with no directly relevant source section, a test of whether general political context transfers to a specific coding task, a comparison of AI MAE against V-Dem's published IRT-corrected country-year scores alongside the primary panel-mean target, to check for a systematic gap between the two that would suggest fine-tuning absorbs biases specific to raw coder ratings, and early feasibility checks on persona framing and decoding temperature, intended to establish groundwork for a possible follow-on study rather than to answer a question posed by this one.